Downloaded from gsabulletin.gsapubs.org on June 29, 2014

Geological Society of America Bulletin

Magnetic polarity stratigraphy and palynostratigraphy of the Mississippian-Pennsylvanian boundary interval in eastern North America and the age of the beginning of the Kiaman

N.D. Opdyke, P.S. Giles and J. Utting

Geological Society of America Bulletin 2014;126, no. 7-8;1068-1083
doi: 10.1130/B30953.1

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

© 2014 Geological Society of America



THE
GEOLOGICAL
SOCIETY
OF AMERICA

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

Magnetic polarity stratigraphy and palynostratigraphy of the Mississippian-Pennsylvanian boundary interval in eastern North America and the age of the beginning of the Kiaman

N.D. Opdyke[1], P.S. Giles[2,†], and J. Utting[3]
[1]*Department of Earth Sciences, University of Florida, Gainesville, Florida 32611, USA*
[2]*Natural Resources Canada, Geological Survey of Canada (Atlantic), P.O. Box 1006, Dartmouth, B2Y 4A2 Nova Scotia, Canada*
[3]*Natural Resources Canada, Geological Survey of Canada (Calgary), Calgary, T2L 1A7 Alberta, Canada*

ABSTRACT

Magnetic polarity data from the Maritimes Basin in eastern Canada reveal patterns of magnetic reversal through ~17 m.y. of late Mississippian and early Pennsylvanian time. We collected samples from three sections on Cape Breton Island, a fourth that extends a section studied previously, and a fifth from a deep cored well on Prince Edward Island. We compared the polarity data from these sections to previous results from lower parts of the Joggins section of eastern Canada. All sections are constrained by new palynological data and collectively span the late Mississippian, Brigantian, Pendleian, and much of the Arnsbergian substages. A major unconformity at the Pennsylvanian-Mississippian boundary makes correlation difficult. The late Mississippian Arnsbergian substage is dominated by normal polarity, whereas the Pendleian and Brigantian are characterized by normal and reversed polarity intervals in about equal proportions. In the United States, the Mississippian-Pennsylvanian boundary is a major hiatus that juxtaposes strata of early Pennsylvanian age on Mississippian strata ranging in age from Brigantian to mid- to late Arnsbergian. Data for the Mauch Chunk Formation in the United States provide polarity reversal patterns for correlative sequences in the Central Appalachian Basin. Field mapping and magnetic polarity data show closely comparable Mississippian-Pennsylvanian relationships in the Maritimes Basin more than 1500 km northeast from the U.S. sections. The eustatic event that coincides with the Mississippian-Pennsylvanian boundary unconformity is late Arnsbergian to Yeadonian in both the central Appalachians and Maritime Basins. Basal Pennsylvanian strata of Yeadonian to Langsettian age on Cape Breton Island reveal normal paleomagnetic polarity through several hundred meters of section. These strata are overlain by a thick section showing only reversed polarity. We believe that this polarity boundary is the base of the Kiaman. An early Langsettian position (ca. 318 Ma) for the base-Kiaman superchron is indicated by these new data. In eastern Nova Scotia and New Brunswick, the sedimentary rocks just above the unconformity are of reversed polarity. It is clear that the sedimentation in the two areas is affected by the unconformity, and no magnetic or palynological correlation is possible. Above the unconformity, only *saetosa* zone sediments are present.

INTRODUCTION

The sedimentary fill of the Maritimes Basin of eastern Canada (Fig. 1) consists of more than 10 km of predominantly continental sedimentary rocks with abundant, fine-grained red beds ranging in age from mid-Devonian to Early Permian (Gibling et al., 2009), and it is thus an ideal region in which to pursue late Paleozoic polarity magnetostratigraphy. These red beds have been of interest to paleomagnetists since the 1960s. In addition to the magnetic stratigraphy reported by DiVenere and Opdyke (1990, 1991b), other researchers have noted both normal and reverse polarities in rocks ranging from Tournaisian to Artinskian (Early Permian) in age (Roy and Robertson, 1968; Roy and Park, 1969; Symons, 1990; Buchan and Chandler, 1999), although none of these studies involved close sampling and biostratigraphically defined stratigraphic sections. The study reported here seeks to extend these early studies and refine the polarity magnetostratigraphic correlations within eastern North America as well as to determine the position of the beginning of the Kiaman.

Earth's magnetic field consists of two states, one in which the magnetic field reverses at a rate of ~3 or more times per million years and another in which it does not reverse at all (Opdyke and Channell, 1996). The record of the magnetic field shows that there are at least two intervals of very few or no reversals of the magnetic field, and there may be more. The most recent of these intervals is the Cretaceous Quit zone, and the other, in the late Paleozoic, is called the Kiaman, named by Irving and Parry (1963) from lavas at Kiama point in New South Wales, Australia. The beginning of the Kiaman has not been placed in the stratigraphic record and is the focus of this study. Why Earth's magnetic field has these two states is unknown at this time.

Late Mississippian–Pennsylvanian Stratigraphic Overview

The Maritimes Basin is a rather general term applied to widely distributed middle Devonian to Early Permian sedimentary rocks in eastern Canada. Regionally, three large structural basins, identified respectively as the Magdalen, Sydney, and St. Anthony's Basins, comprise the so-called Maritimes Basin (Fig. 1).

Fine-grained lacustrine gray mudstones and terrestrial red beds of the early Namurian (Serpukhovian) Mabou Group lie conformably on latest Viséan marine rocks in central and eastern Nova Scotia and are mid- to possibly late Arnsbergian in age where preserved immediately beneath the Mississippian-Pennsylvanian unconformity (Fig. 2). In those parts of our study area where the upper part of the Windsor Group retains its complete record of marine sedimentation, the Hastings and Pomquet Formations comprise the Mabou Group (Belt, 1964, 1965; Boehner, 1985). With its type section on the eastern shores of the Strait of Canso, where it is exposed south of the village of Hastings on Cape Breton Island, gray shales of the Hastings

†E-mail: pgiles@nrcan.gc.ca.

GSA Bulletin; July/August 2014; v. 126; no. 7/8; p. 1068–1083; doi: 10.1130/B30953.1; 12 figures; Data Repository item 2014139; published online 2 May 2014.

For permission to copy, contact editing@geosociety.org
© 2014 Geological Society of America

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

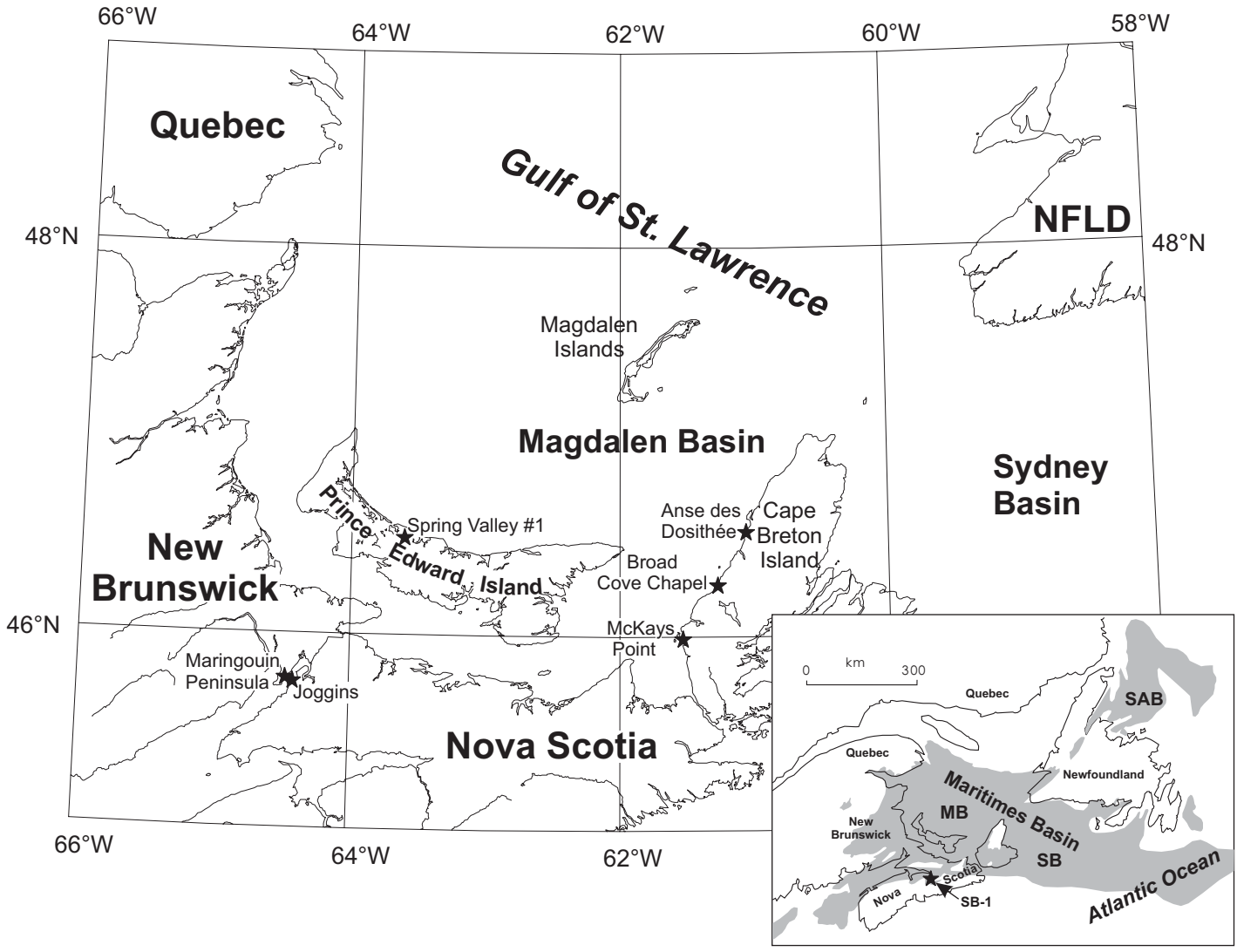


Figure 1. Location map of the studied sections in the southwestern parts of the Magdalen Basin (MB) in Nova Scotia, Prince Edward Island, and southeastern New Brunswick. NFLD—Newfoundland; MB—Magdalen Basin; SB—Sydney Basin; SAB—St. Anthony's Basin.

Formation have been mapped throughout the Antigonish-Mabou subbasin of Wade et al. (1977) by Boehner and Giles (1982) and Giles et al. (1997a, 1997b, 1997c). The Hastings and Pomquet Formations are also widely distributed in southwestern Cape Breton Island (Hamblin, 2001).

The Pomquet Formation is named for exposures in its type section on the Pomquet River south and east of the town of Antigonish in northern mainland Nova Scotia. With a regional distribution similar to that of the Hastings Formation, the Pomquet is conformable and transitional with the Hastings, and of comparable thickness at 300–500 m. Fine-grained red beds characterize the Pomquet Formation, the age of which is Namurian A (Neves and Belt, 1971) and Arnsbergian (Utting and Giles, 2008). The Pomquet Formation is nonmarine, deposited on a low-relief, low-slope floodplain (Hamblin, 2001), and its upper boundary is the Mississippian-Pennsylvanian unconformity.

The Mabou Group in the Nova Scotia–New Brunswick border region consists of, in ascending order, the (Middleborough =) Maringouin Formation, the Shepody Formation, and the (Claremont =) Enragé Formations. The Maringouin Formation is dominated by massive red mudrocks with intercalated very fine-grained red sandstones, and it overlies the fossiliferous Lime Kiln Brook Formation (middle Windsor Group) of late Asbian age (von Bitter et al., 2006). Sandstone is more abundant in the upper parts where the Maringouin Formation passes transitionally upward into the sandstone-dominated Shepody Formation (St. Peter and Johnson, 2009). In its lower parts, the Shepody Formation is composed of relatively thick channel sandstone bodies with intervening red siltstone and mudstone intervals, which can reach 50 m in thickness. The upper Shepody Formation is typified by multistoried gray sandstones in thick channel bodies, intercalated with mainly gray mudrocks (St. Peter and Johnson, 2009). Plant remains are ubiquitous in the thick sandstones. This part of the formation is entirely unsuited for paleomagnetic study.

The Enragé Formation, with a typical brick-red color, conformably and transitionally overlies the Shepody Formation. Arkosic sandstones and coarse grits characterize this formation. A fining-upward motif is common, in repeated cycles 4 to 10 m in thickness. The upper boundary with the Boss Point Formation is concor-

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

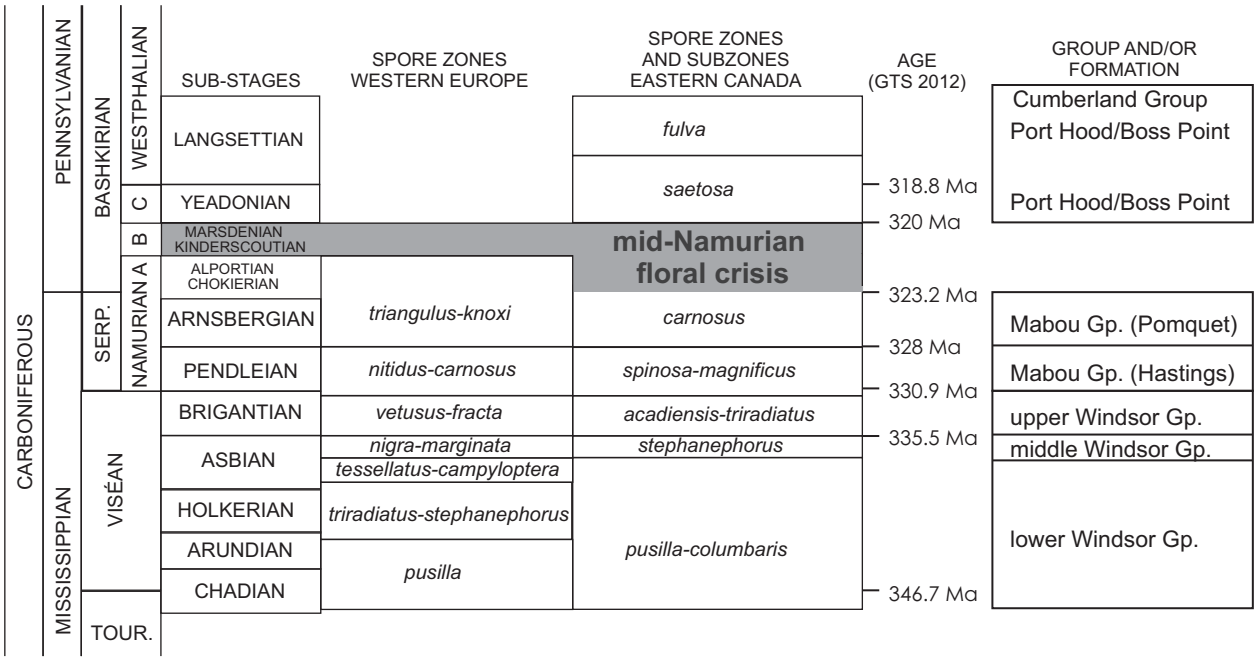


Figure 2. Carboniferous spore zones recognized in western Europe and in eastern Canada. Stratigraphic units are shown on the right.

dant, but it represents a significant hiatus and is here interpreted as a disconformity.

Earliest Pennsylvanian strata in the Magdalen Basin are typified by multistoried fluviatile sandstones, interbedded with gray or red mudrocks. In northwestern Nova Scotia and southeastern New Brunswick, these rocks are assigned to the Boss Point Formation. In western Cape Breton Island, they are termed the Port Hood Formation. Their age is latest Namurian (Yeadonian) to Westphalian A (Langsettian) as determined from macroflora (Utting and Wagner, 2005) and from palynology (Utting et al., 2005, 2010). Collectively, these early Pennsylvanian rocks comprise the lower part of the Cumberland Group (Ryan and Boehner, 1994). In the lower parts of the Boss Point Formation, thin gray mudrocks separate the channel sandstones. In the Cape Breton Island sections, overbank fines in the lower part of the Port Hood Formation (Margaree Member) are almost entirely red in color in intervals that may measure tens of meters in thickness but are typically thinner. The Margaree Member of the Port Hood Formation thus compares in many aspects with the Little River Formation in the Joggins section, which transitionally overlies the type Boss Point Formation and is characterized by fine-grained red beds with channel sandstones (Calder et al., 2005).

The lower Cumberland Group contains coal beds. In western Cape Breton Island, these strata are assigned to the Colindale Member of the Port Hood Formation, conformably overlying the Margaree Member. In the Nova Scotia–New Brunswick border region, equivalent coal measures comprise the Joggins and Springhill Mines Formations. Historically, these coals have been an important local resource both at Joggins and at Port Hood.

PALYNOSTRATIGRAPHY

Palynological samples were treated with acid to separate the pollen which were then observed by microscope and identified. The data are archived in Calgary by the Geological Survey of Canada (GSC), and the position of these samples is indicated on the stratigraphic sections.

We sampled rock units of late Mississippian to earliest Pennsylvanian age and assigned them to microfloral zones following Utting and Giles (2004, 2008), von Bitter et al. (2006), and Utting et al. (2010) (Fig. 2). The floral *spinosa-magnificus* zonal indicators occur in the lowest sample of the Shepody Formation in its type section on the Maringouin Peninsula (Utting et al., 2010). Claremont-equivalent Enragé Formation strata in New Brunswick are exposed immediately beneath the Mississippian-Pennsylvanian unconformity, but palynological dating of the Enragé Formation is not definitive. Thus, in the Nova Scotia–New Brunswick border region sections, we are presently unable to confirm the presence of the latest Mississippian *carnosus* zone in any section and are therefore unable to rigorously place the top of the *spinosa-magnificus* zone. The strata of the Boss Point Formation at the base of the Pennsylvanian are thought to be latest Namurian (Yeadonian) to Westphalian A (Langsettian) in age (Utting et al., 2010). They lack many taxa characteristic of the late Mississippian and postdate the mid-Namurian floral crisis (Fig. 2), which has been called the Gothan Floral Break by Menning et al. (2006).

The Broad Cove Chapel section on Cape Breton Island includes all of the *spinosa-magnificus* zone and a significant part of the overlying *carnosus* zone of Utting and Giles (2008). In those exposures, the collection of new paleomagnetic data started immediately above the highest marine carbonate interval at the top of the Windsor Group. Near the base of this 10-m-thick carbonate unit, the first *spinosa-magnificus* zonal indicators were recovered in the uppermost part of the Windsor Group. Thus, the *acadiensis-triradiatus* to *spinosa-magnificus* zonal boundary coincides approximately with the Windsor-Mabou Group boundary, placed by Giles et al. (1997b) at the top of the of this limestone. At the approximate level of the Hasting-Pomquet Formation boundary in this section, the first *carnosus* zone indicator palynomorphs appear, and they continue up section to the exposed Mississippian-Pennsylvanian unconformity. At its base, the *carnosus* zone is questionably Pendleian in age, but the zone is mainly of Arnsbergian age (Utting and Giles, 2008). The *carnosus* zone terminates at a regional unconformity at the top of the Mississippian, and its top cannot therefore be rigorously defined with new incoming taxa and has been designated an assemblage zone (Utting and Giles, 2008).

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

Beds immediately above the Mississippian-Pennsylvanian unconformity are distinctly different palynostratigraphically from older strata in eastern Canada, lacking substantial numbers of taxa widely distributed in the preceding *acadiensis-triradiatus* zone (Utting et al., 2010). In western Cape Breton Island, this change in palynoflora coincides with the base of the Port Hood Formation; in the Joggins section, the change occurs at the base of the Boss Point Formation. In each case, the change marks the first preservation of Pennsylvanian strata. Our assessment of age for these earliest Pennsylvanian beds is late Yeadonian to Langsettian, based on macrofloral assessment by Utting and Wagner (2005), and on palynological assessment by Utting et al. (2010). Thus, the Mississippian-Pennsylvanian unconformity in western Cape Breton Island falls within an interval extending from late (but not latest) Arnsbergian to the late Yeadonian. No rock record has yet been documented within this time interval in the Maritimes Basin, but it is represented in western Europe by the Chokierian, Alportian, Kinderscoutian, and Marsdenian substages. The Mississippian-Pennsylvanian boundary in the Maritimes Basin is a major regional unconformity, as it is in the Central Appalachian Basin in the United States (Blake, 2006; Maynard et al., 2006; Blake and Beuthin, 2008), and it was probably caused by eustatic lowering of sea level as a result of initiation of the glaciation of Gondwana.

Mississippian portions of the Cape Breton Island sections at McKays Point and at Anse des Dosithée fall entirely within the *carnosus* zone, based on the appearance in our lowest samples of *Savitrisporites nux* and *Crassispora kosankei*. Like the Broad Cove section, each of these two sections in its upper parts crosses the Mississippian-Pennsylvanian unconformity, and comparable palynomorph assemblages are found in all three Cape Breton sections, respectively, above and beneath the Mississippian-Pennsylvanian unconformity.

An additional short section included in this study represents the entire Mississippian section (~86 m) cored in the Spring Valley well (Giles and Utting, 1999) on Prince Edward Island, which is overlain by Lower Pennsylvanian strata ~360 m in thickness. The Mississippian spore assemblages are dominated by simple trilete spores, e.g., *Calamospora parva*, *Punctatisporites glaber*, *Leiotriletes* sp., *Lophotriletes* sp., and *Verrucosisporites* sp. A common species is *Schopfipollenites acadiensis*, and *Knoxisporites triradiatus* is present, which suggests an age no older than the Brigantian. However, the presence of *Crassispora kosankei* in these three samples suggests an age equivalent to the *carnosus* zone of Arnsbergian age. Palynological allocation to the *carnosus* zone is consistent with our reassignment of these rocks to the Enragé Formation.

Overlying rocks in the Spring Valley well, beginning at a depth of 1620 m, contain early Pennsylvanian palynomorph assemblages characteristic of the *saetosa* zone of Utting et al. (2010) and are dominated by *Potonieisporites elegans*, *Schopfipollenites ellipsoids*, and members of the *Potonieisporites-Florinites-Wilsonites-Guthoerlisporites* monosaccate complex. At a depth of 1520 m, the *fulva* zone makes its first appearance, with *Raistrickia fulva*, *Laevigatosporites vulgaris*, and *Laevigatosporites minor.* This suggests that the earliest Pennsylvanian *saetosa* zone is represented by only a limited thickness in the Spring Valley well, and that the overlying *fulva* zone is truncated by the "Westphalian unconformity" (Giles and Utting, 1999; see following). The limited representation of the *saetosa* zone suggests that Pennsylvanian deposition on Prince Edward Island began later than deposition at Joggins or in western Cape Breton Island. At 1256 m, *Vestispora magna* appears, along with *Vestispora tortuosa*, *Vestispora costata*, and *Vestispora fenestrata*. The simultaneous first appearance of these four taxa suggests a hiatus and assignment to the *Vestispora magna* zone of Utting et al. (2010) of Duckmantian and probable late Duckmantian age. We suggest that these beds represent the first deposition above a significant hiatus spanning the latest Langsettian and early Duckmantian in this well.

Utting et al. (2010) documented the presence of the *saetosa* and *fulva* zones in the Joggins succession, appearing first in the oldest Pennsylvanian Boss Point Formation. The *saetosa* zone in the Joggins section is ~1400 m in thickness. The first occurrence of *R. fulva* is near the base of the recently amended Joggins Formation (Davies et al., 2005). In western Cape Breton Island, we sampled the Port Hood Formation at Broad Cove Chapel, where the Pennsylvanian interval sampled is more than 800 m thick. There, the entire Pennsylvanian succession contains palynomorphs of the *saetosa* zone in strata typical of the lower Port Hood Formation. *Saetosa* zone assemblages characterize the short Pennsylvanian successions at McKays Point and Anse des Dosithée as well.

MAGNETIC POLARITY STRATIGRAPHY

Methods and Techniques

This study integrates the stratigraphy, palynology, and magnetic polarity stratigraphy of sections in the middle Carboniferous of Nova Scotia and New Brunswick. All paleomagnetic and palynological samples were placed in stratigraphic position in published sections or sections measured by the authors. The palynological sites were taken from unoxidized sediments often from thin or very thin beds. Paleomagnetic samples were drilled from red siltstones or sandstones and positioned in the stratigraphy. Paleomagnetic sampling was carried out in well-exposed coastal sections using a handheld gasoline-powered diamond-core drill. The resulting cores were 2.5 cm in diameter and varied from 2.5 cm to 5 cm in length. Orientation of samples was measured with a Brunton geological field compass, attached to a Pomeroy orienting device. Samples were taken mainly in fine-grained red sandstone and siltstone beds. The cores were sliced to a length of ~2.5 cm (1 in.) and demagnetized thermally, using a Mu-metal shielded oven. The measurements of remanent intensity and direction were made in a cryogenic magnetometer housed in a Mu-metal shielded room. All samples in this study were thermally demagnetized. The oven temperature was increased incrementally in 12 or more steps to temperatures as high as 700 °C; after each step, the samples were remeasured (Fig. 3). The temperature interval between measurements was decreased at higher temperatures. If no decay to the origin was seen, the result was not deemed successful, and no virtual geomagnetic pole (VGP) was calculated. Polarity could often be determined from the demagnetization plots when demagnetization trajectories bypassed the origin. The VGP for each successfully demagnetized sample was calculated and plotted for each sample (site) against the paleolatitude determined from the paleomagnetic pole (Fisher, 1953) calculated from the data (see GSA Data Repository[1]). Unreliability in the magnetic data is seen as VGP values plotting at low latitude. Normal directions are directed to the north in this study.

The behavior of the samples in this study during thermal demagnetization was essentially the same as previously reported by DiVenere and Opdyke (1990). All components of magnetization were determined following techniques described by Kirschvink (1980). Three separate components have been previously isolated in these rocks and have been termed component A (Kiaman, exclusively postfolding reverse directions), component B (synfolding, acquired during folding), and component C (prefolding) and are indicated by the dashed lines in Figure 3

[1]GSA Data Repository item 2014139, geologic and geographic situations of samples as well as the demagnetization procedure, is available at http://www.geosociety.org/pubs/ft2014.htm or by request to editing@geosociety.org.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

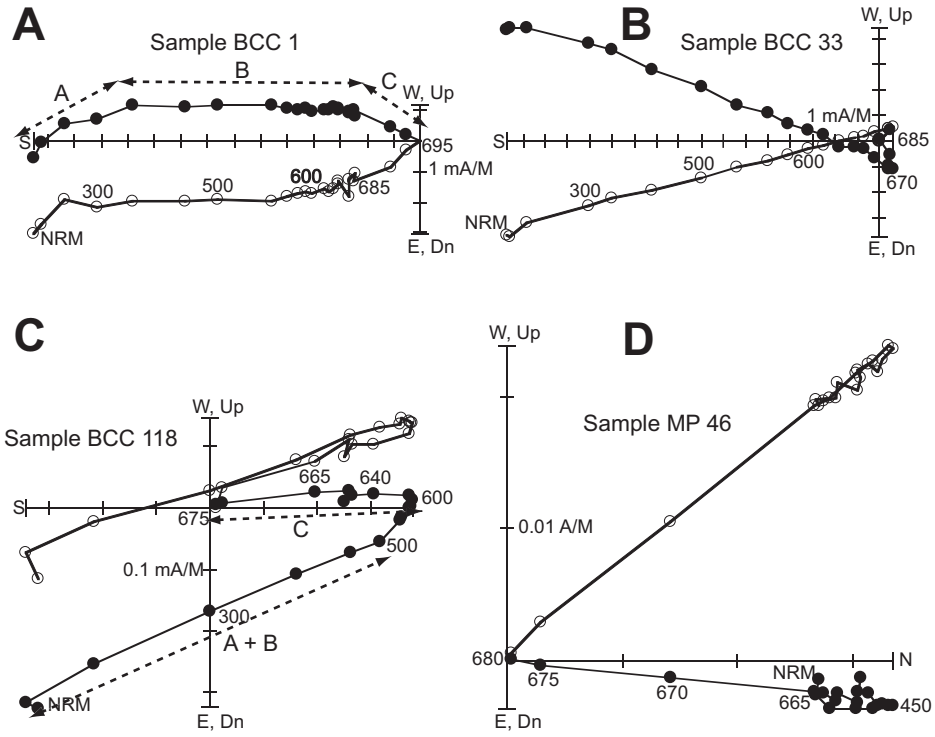


Figure 3. (A, C) Examples of thermal demagnetization behavior by specimens from four samples from red beds examined in this study showing two or more magnetization components. (B) A sample with (A + B) component that obscures the C component, which is not isolated even at high temperatures. (D) The resolution of a "C" component relatively unaffected by overprinting. NRM—natural remanent magnetization.

for normal and reverse directions. These are isolated at increasingly higher unblocking temperatures (Fig. 3).

It is now known that magnetic results from many red beds have suffered from inclination shallowing (King, 1955; Tauxe and Kent, 1984; Kodama, 2012). Tauxe and Kent (2004) have produced a magnetic field model that allows a correction to be made for this effect. This correction was applied to our data and results in an inclination of the paleomagnetic field of 29.5°, which is an increase of ~10° from that measured (Fig. 4). This inclination would place the sediments at ~15°S latitude, in an arid setting. The explanation of this correction is found in Tauxe and Kent (2004). It should be noted that the latitude of the sediments as they are deposited changed from a south latitude setting to very nearly an equatorial position at the time of folding. This changed the climatic setting of the depositional basins and ultimately led to deposition of extensive coal measures (Bilardello and Kodama, 2009).

Positive fold tests for the "C" component have been obtained in the Joggins and Maringouin Peninsula sections and were reported in the earlier work of Di Venere and Opdyke (1990, 1991b). The mean "C" directions for the four sections presented in this study were combined, and a paleomagnetic pole was calculated using Fisher (1953) statistics. The mean directions, with the α_{95} circle of confidence, are plotted in Figure 5. The mean directions are shown before and after bedding tilt correction. A dramatic change is seen in the concentration of the directions with k (the concentration parameter) increasing from 4.8 to 159. It is clear that the "C" component is prefolding. Therefore, we can use the results to create a magnetic stratigraphy.

VGP values were calculated for each sampling level, and the latitude of the VGP was determined with respect to the paleomagnetic pole for these sediments. The VGP latitudes were then used to construct the magnetic stratigraphy. All paleomagnetic results have been deposited in the GSA Data Repository (see footnote 1).

Combined Stratigraphy

Lithostratigraphic control for both palynological and paleomagnetic sampling at Broad Cove Chapel (Fig. 6) and Anse des Dosithée (Fig. 7) was provided by measured sections published by Keighley and Pickerill (1996). The section at McKays Point was measured and described (by Giles) during sampling for this study (Fig. 7). All samples were plotted adjacent to a summary graphic log for each section, in their appropriate relative positions as determined in the field, or by depth for the Spring Valley well. The basal part of the Spring Valley well (Fig. 8) is presented graphically from descriptions by P.S. Giles. Palynological and paleomagnetic sample positions in the Enragé and Boss Point Formations in the coastal section near Green Creek in southeastern New Brunswick were located stratigraphically using an unpublished log of that section kindly provided by Clinton St. Peter and Susan Johnson of the New Brunswick Department of Natural Resources.

Spring Valley Well

Subsampling of diamond-drill cores from the basal part of the Spring Valley Number 1 well from sections 15 cm in diameter was done with the same gasoline-powered core sampling drill that was used in outcrop. Orientation of the core was accomplished using the well-documented eastern Canadian Kiaman overprint (180°) as a directional guide and following procedures outlined by Kent et al. (1995). The magnetic directions for Spring Valley samples were rotated to the overprint direction, and the VGP was determined from the rotated direction. Primary magnetic remanence for the Spring Valley well samples could not be rigorously demonstrated because the original cores could not be securely and independently oriented, and the beds are not folded. However, using the regional Kiaman overprint for sample orientation, our data indicate that normal polarity is dominant in the lower 48 m of the core (Mississippian) and is followed by reverse polarity continuously to the level of the Mississippian-Pennsylvanian boundary. Earliest Pennsylvanian rocks contain palynomorphs of the *saetosa* zone (Fig. 8), presumed here to be high in that zone, overlain by the *fulva* zone. These rocks are reversely magnetized, as are all Pennsylvanian strata in the Spring Valley well. In the Joggins shore section, rocks bearing *saetosa* and *fulva* zonal assemblages comprise the Little River Formation (formerly considered upper Boss Point Formation) and the Joggins Formation, respectively. Although magnetic polarity within the Joggins Formation has not been determined, the Little River Formation is entirely reverse in polarity (DiVenere and Opdyke, 1990, 1991b). Thus, results from the Spring Valley well for rocks of comparable biostratigraphic position are of reverse magnetic polarity as well.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

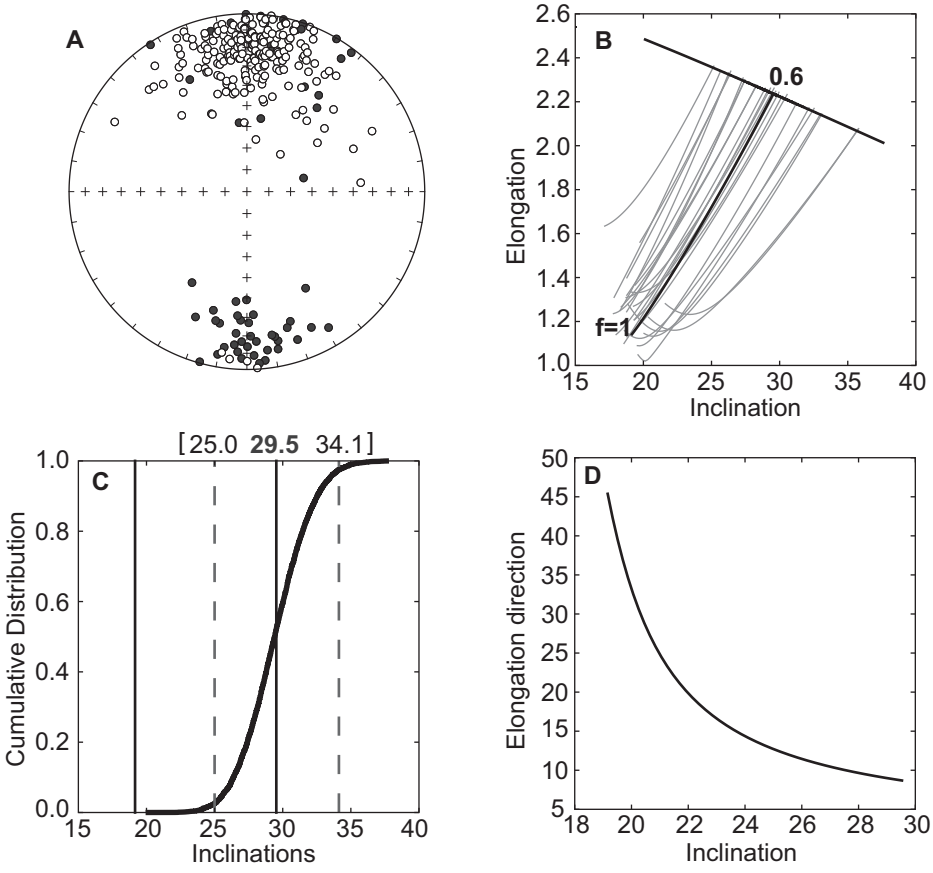


Figure 4. Site mean directions for the "C" component. (A) Plotted on an equal-area projection, the directions below (above) the horizontal plane are indicated by filled (open) circles. (B) Plot of inclination vs. elongation of the data set, determined using the bootstrap method. (C) Plots of inclinations against the cumulative distribution resulting in a correction for inclination error of 10°, giving a corrected inclination of 29.5° with error limits of 25° to 34°. (D) Plots of elongation direction against inclination.

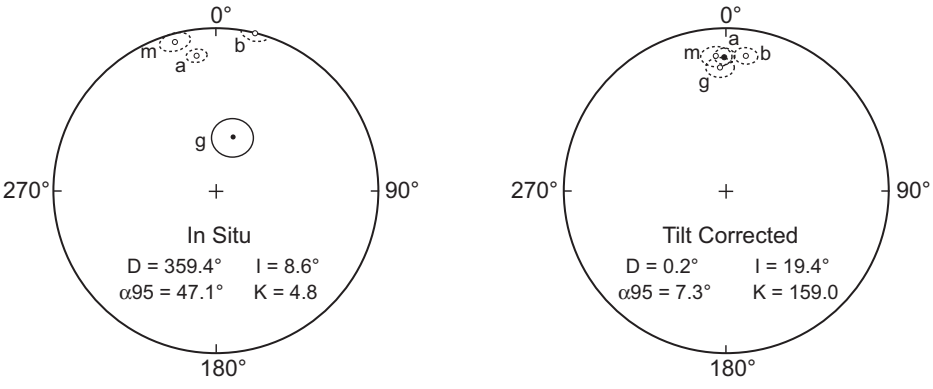


Figure 5. Successful fold test using paleomagnetic data (section grand means) from western Cape Breton Island sections and from the Green Creek section, southeastern New Brunswick. a—Anse des Dosithée, b—Broad Cove Chapel, m—McKays Point; g—Green Creek section. Statistics before and after correction for bedding tilt are given on the equal-area projections.

SB1—Windsor Group, Shubenacadie Basin, Southern Nova Scotia

SB1 is a cored mineral exploration well drilled near the town of Stewiacke on the Meguma Platform of southern Nova Scotia, which was previously sampled for palynology (Utting, 1980). The well was cored continuously from surface to a depth of ~915 m (~3000 ft) and is designated the principal reference section for the Windsor Group. Red beds throughout the middle and upper parts of the Windsor Group were sampled for magnetic polarity determination, so that the polarity stratigraphy of this biostratigraphically well-defined polarity succession could be compared with the equivalent, but undated Middleborough and Maringouin Formations of the Nova Scotia–New Brunswick border region. Unfortunately, all SB1 samples displayed viscous behavior during demagnetization, and the results were not interpretable. The results are reported here for the sake of completeness.

Western Cape Breton Island

The Mississippian section exposed at Broad Cove Chapel (Fig. 6) is more than 600 m thick, extending from the upper part of the Windsor Group through the Hastings and Pomquet Formations of the Mabou Group. Samples were taken through the upper 500 m of the Mississippian section, spanning the entire exposed thickness of the Mabou Group. The Hastings Formation, dominated by gray and dark-gray shales, could not be sampled in comparable detail to the red bed–dominated Pomquet Formation. In ascending order, the section records ~40 m of reverse polarity at the base, overlain by 80 m of normal polarity. At this level, a gap in the sample record is ~75 m thick, above which normal polarity is again characteristic, continuing to the top of the Mississippian, immediately beneath the Pennsylvanian unconformity. Thus, our data show a single polarity reversal from reverse to normal near the base of this section, within the lower part of the *spinosa-magnificus* zone, and we correlate this reversal to the base of N5 in the Joggins section (Di Venere and Opdyke, 1991b) (Fig. 9).

The Pennsylvanian section sampled at Broad Cove Chapel is continuously exposed above the underlying Mississippian succession and totals nearly 1100 m in thickness. We sampled ~850 m of this section, beginning immediately above the Mississippian-Pennsylvanian unconformity. Five samples from the lowest 22 m of this Pennsylvanian section all yielded normal polarity. At this level, an absence of suitable lithology within the drab sandstones of the lower Port Hood Formation results in a 140 m gap in sampling. Above this gap, sampling

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

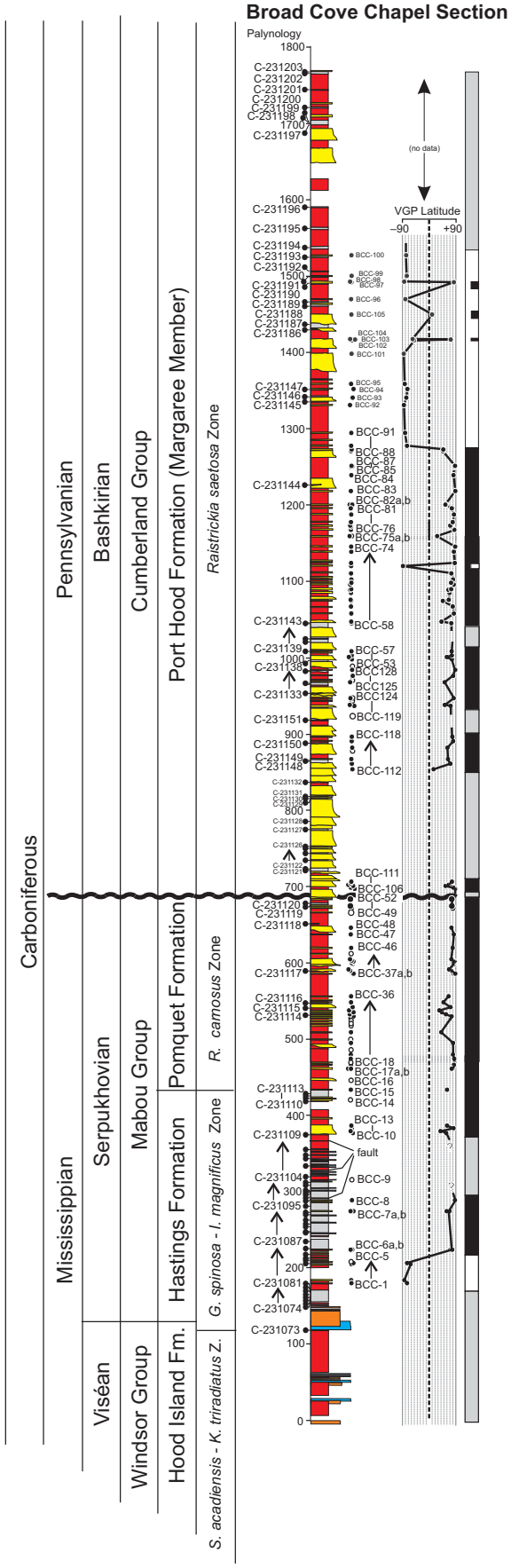


Figure 6. Distribution of palynological and paleomagnetic samples and magnetic polarity stratigraphy in the section at Broad Cove Chapel, Cape Breton Island. Vertical scale in meters. Black and very dark-gray shades identify Windsor Group carbonate rocks and gypsum, respectively. Black indicates normal polarity, white reverse.

426 m of section yielded normal polarity in all but one (reversed polarity) sample, which was located ~428 m above the Mississippian-Pennsylvanian unconformity (Fig. 6). The top of this thick normal zone is defined by an abrupt change to reverse polarity, closely constrained by sample spacing, and the section continues for 257 m with all but two samples having reverse polarities. Single samples at 730 m and 804 m above the basal Mississippian-Pennsylvanian unconformity yielded normal polarities. We follow convention and do not define magnetozones on the basis of these single-sample (site) results. Overall, the Pennsylvanian section at Broad Cove Chapel contains two thick magnetozones, the lower of normal polarity, and the upper reverse. This change in polarity probably represents the Kiaman boundary because all younger rocks have reverse polarity.

At McKays Point, only normal polarity was observed through a 400-m-thick section that represents a part of the Mississippian Pomquet Formation and extends nearly 50 m into the overlying Port Hood Formation. Polarity through a 30 m sampling gap spanning the Mississippian-Pennsylvanian unconformity could not be determined, although normal polarity was found in three samples of Port Hood Formation (Pennsylvanian) red beds above this gap (Fig. 7).

At Anse des Dosithée, samples were taken through 335 m of Mississippian strata and 173 m of immediately overlying Pennsylvanian strata. Mississippian rocks belong entirely to the Pomquet Formation and are assigned to the *carnosus* assemblage zone of Pendleian and Arnsbergian age. All Mississippian samples are of normal polarity. Lower Port Hood Formation sandstones and shales comprise the earliest Pennsylvanian strata at Anse des Dosithée. All samples from Pennsylvanian strata are of normal polarity, with no large data gaps present. Correlation with the Broad Cove Chapel section suggests that the relatively long sampling gap there, near the base of the Pennsylvanian section, should contain normal polarity rocks.

Southeast New Brunswick Section

On the Maringouin Peninsula, samples for magnetic polarity determination were taken from the upper part of the Shepody Formation, the Enragé Formation, and the Chignecto Bay Member of the lowest Boss Point Formation (Fig. 9). The lowest sampled horizon coincides with the highest sampled horizon of DiVenere and Opdyke (1991b) and thus adds continuous samples through an additional 250 m of section crossing the Mississippian-Pennsylvanian unconformity.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

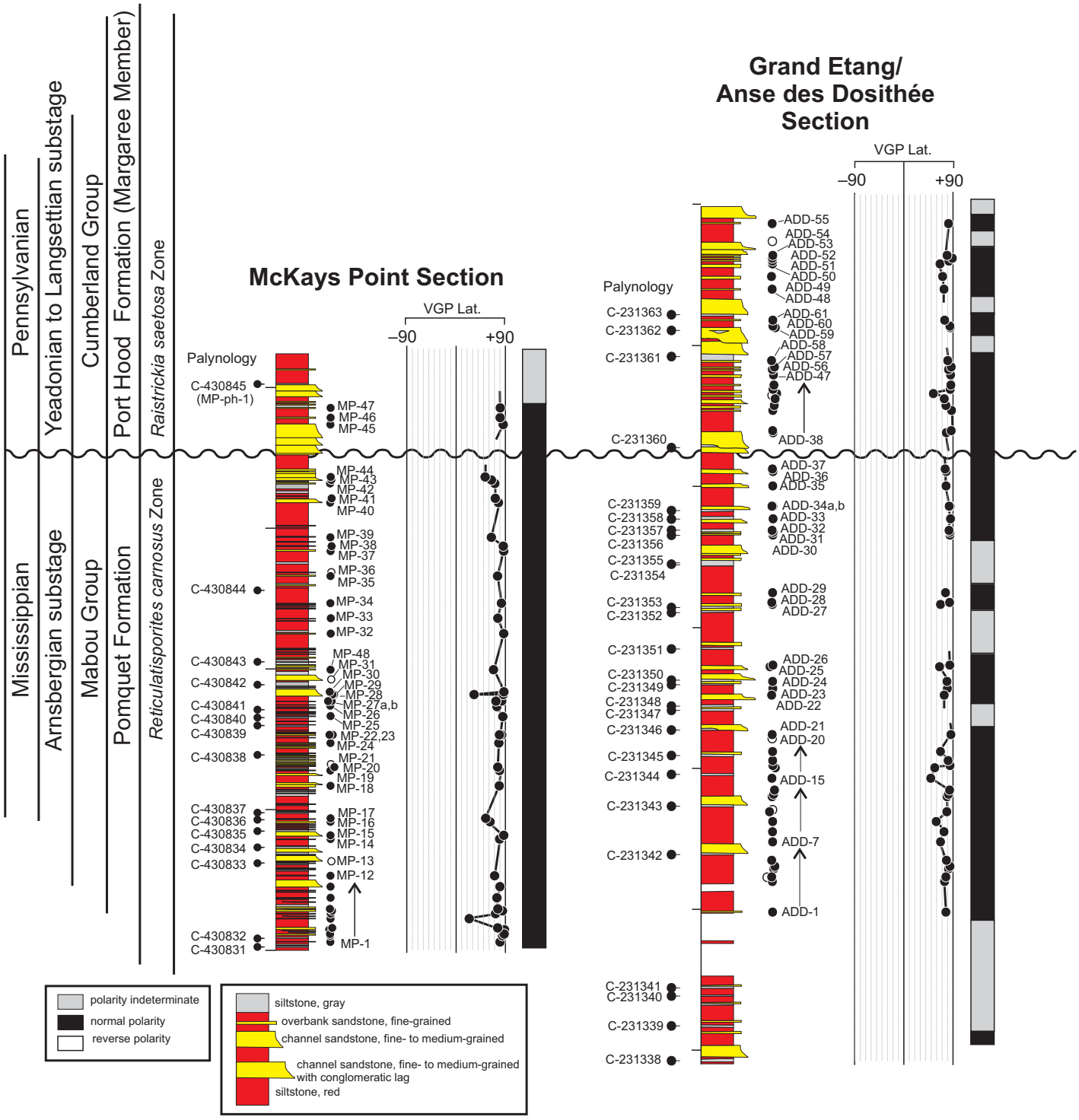


Figure 7. Distribution of palynological and paleomagnetic samples and magnetic polarity stratigraphy at McKays Point, and Anse Des Dosithée. The figure legend and vertical scale are as in Figure 6.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

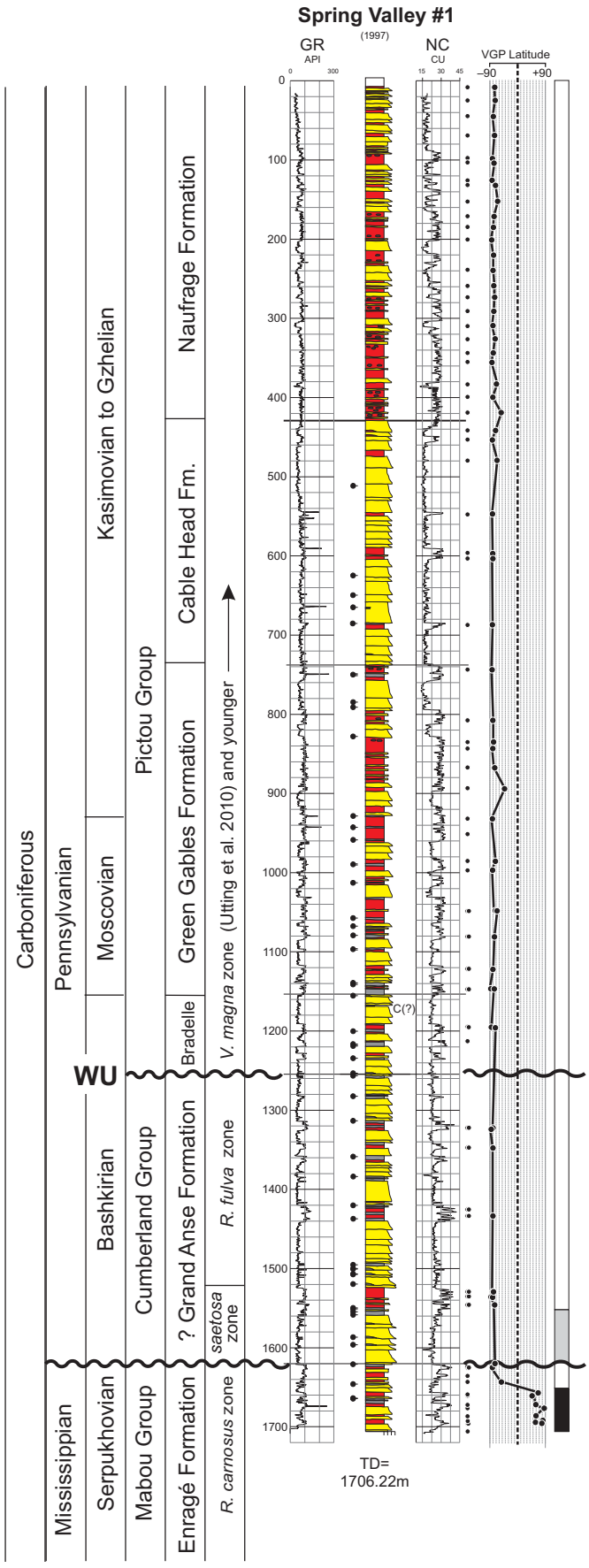


Figure 8. Paleomagnetic polarity stratigraphy of the Spring Valley well plotted against the lithostratigraphy and palynological zonation. Palynological and paleomagnetic legend and vertical scale are as in Figure 6. Plots of the gamma-ray (GR) and electric logs are also given. NC—Neutron Capture log; CU—capture units; API—American Petroleum Institute units for Gamma Ray measurement; WU—Westphalian Unconformity of Giles and Utting (1999); TD—total depth in (m).

The correlations among the new sections on Cape Breton Island and the sections farther to the west on the Maringouin Peninsula and the base of the Joggins section are not obvious. There are some similarities, however. In both regions, the highest strata in the Mississippian have normal polarity. The preserved sediments above the unconformity are of reverse polarity in the Joggins section and on the Maringouin Peninsula but have normal polarity in the three sections on Cape Breton Island. Clearly, the boundary unconformity has played a role in making this correlation difficult. The pollen stratigraphy is of little use since the *saetosa* zone extends from the normal polarity zone at the base of the Pennsylvanian to reverse polarity sediments at the top of the sequence, as demonstrated at Broad Cove Chapel. A possible solution to this problem is to postulate that the normal polarity zone that is low of the Pennsylvanian is missing, caused by erosion or nondeposition in eastern Nova Scotia and the Maringouin Peninsula. Therefore, normal polarity is present low in the Boss Point Formation section. We postulate that the unconformity passes below the red sediments at Boss Point and that the section would correlate to the Broad Cove Chapel section at the base of the Kiaman.

Above the well-defined N5 normal magnetozone of Opdyke and DiVenere (1991), our new results confirm a zone of alternating polarity magnetozones ~150 m in thickness, reaching to within 10 m of the Mississippian-Pennsylvanian unconformity. Polarity data are again well documented ~10 m above the same unconformity; for the highest 90 m of our section, only reversed polarity is found. Unfortunately, the 20 m section spanning the Mississippian-Pennsylvanian unconformity yields no new polarity information. The lowest Boss Point Formation samples, spanning ~100 m entirely within the Chignecto Bay Member, yielded only reverse polarity. This compares with the Boss Point Formation type section, where DiVenere and Opdyke (1991b) found an interval of reversed polarity low in the formation, within which a thin normal polarity magnetozone (N8) was identified. N8 was, until the present work, the highest zone of normal polarity known in the "Namurian" of eastern Canada (Opdyke et al., 2000).

We combined our new magnetic polarity data with previously reported magnetic polarity data and the palynological zonation (Fig. 10), and our correlation of magnetozones between the Nova Scotia–New Brunswick border region and the Cape Breton Island sections is shown in Figure 10. We correlate the position of the *spinosa-magnificus* concurrent range zone, relative to normal magnetozone N5 (DiVenere

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

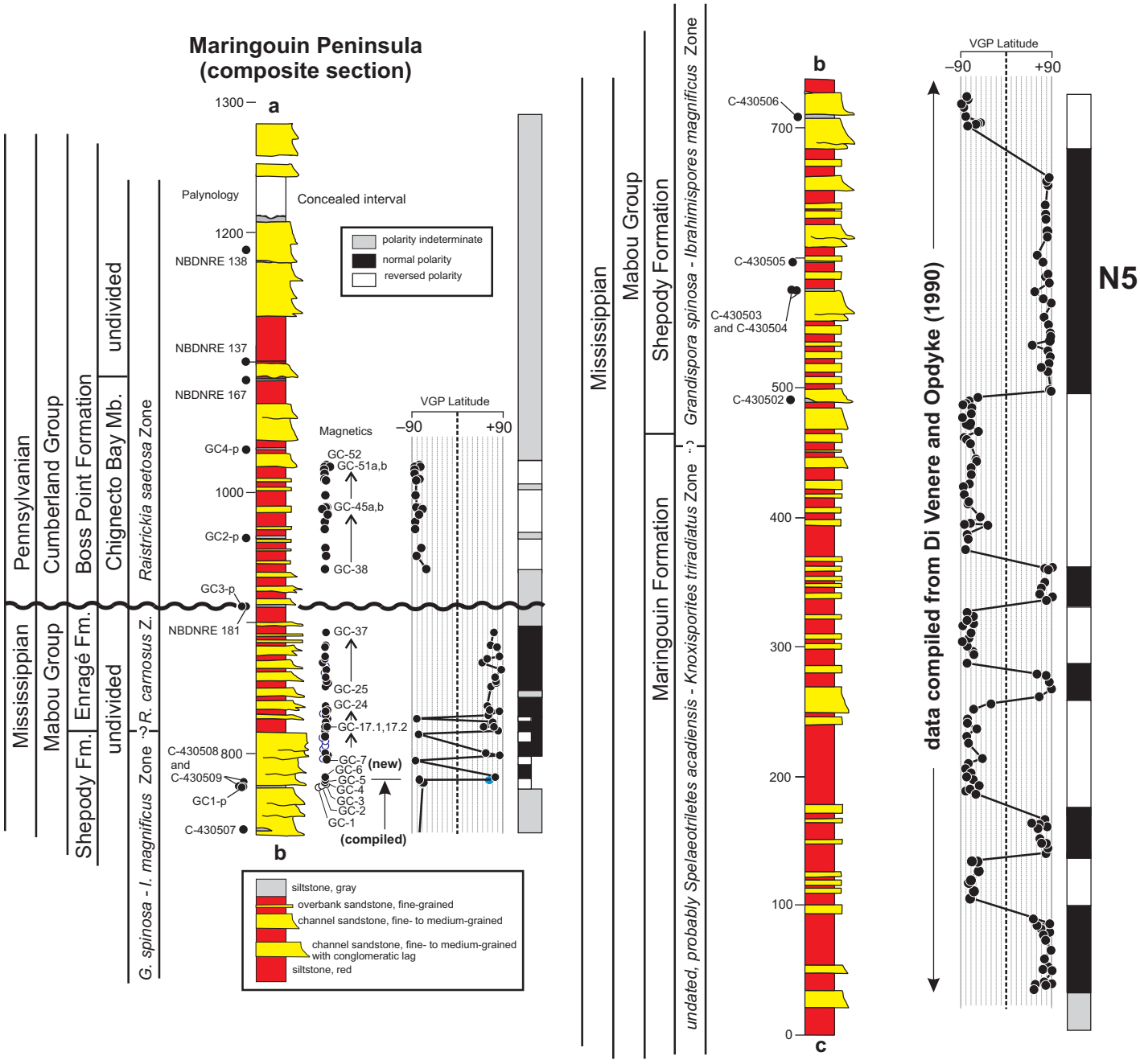


Figure 9. Distribution of palynological and paleomagnetic samples and magnetic polarity stratigraphy in the Maringouin Peninsula composite section, southeastern Nova Scotia (after Di Venere and Opdyke, 1990), based on new data presented in this paper. Virtual geomagnetic poles (VGPs) are plotted against the position of the samples in the measured section. The scale is in meters. Normal polarity is indicated by black, and white indicates reverse polarity.

and Opdyke, 1991b), to the base of the Broad Cove Chapel section in Cape Breton Island at the reverse to normal polarity change just above the Brigantian-Pendleian boundary (Fig. 6). The series of magnetic reversals that typify the late Pendleian and early Arnsbergian at the Joggins section apparently have been removed by the Mississippian-Pennsylvanian unconformity on Cape Breton Island but may fall in sampling gaps at Broad Cove Chapel (Fig. 6). Our data from the western Cape Breton Island sections indicate that the upper parts of the Arnsbergian substage approaching the Mississippian-Pennsylvanian unconformity are entirely of normal polarity.

Our proposed correlation of magnetozones (Fig. 10) is based on the hypothesis that all but the lowest beds of the Claremont Formation at Joggins, and by inference, the correlative Enragé Formation in New Brunswick, should fall within the *carnosus* assemblage zone. The Middleborough and Maringouin Formations, inferred to fall within the Brigantian, remain undated and cannot presently be correlated using polarity stratigraphy into biostratigraphically constrained sections. However, the relationships of these rock units to underlying

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

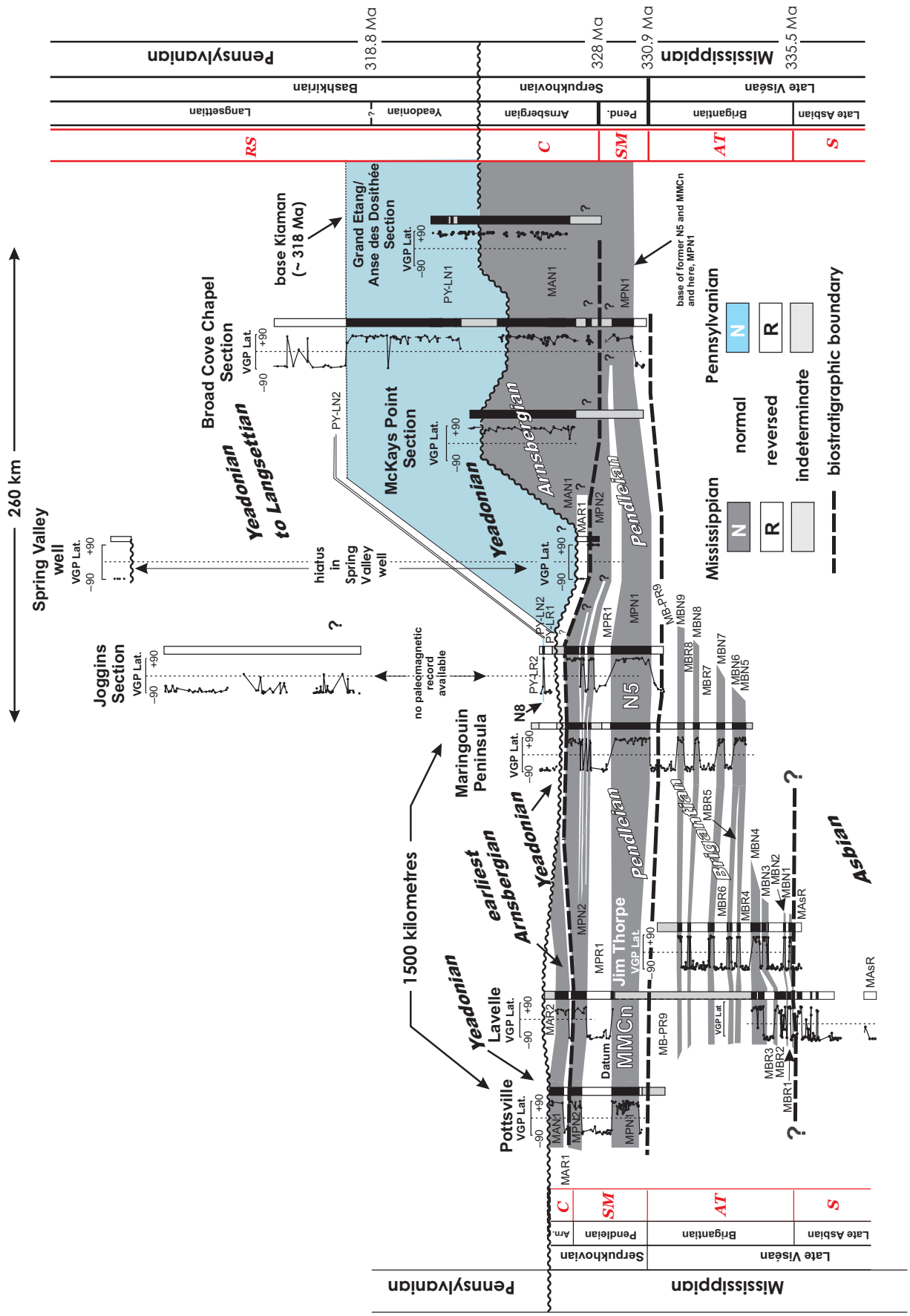


Figure 10. Biostratigraphically supported correlation of late Mississippian and early Pennsylvanian magnetozones from eastern Canada to Pennsylvania. New terminology is proposed for the magnetozones correlated to the palynology and fossil plants. Ages are from the time scale of Gradstein et al. (2012). Unconformities are indicated by wavy lines. RS—*R. saetosa* Zone; C—*carnosus* Zone; SM—*spinosa-magnificus* Zone; AT—*acadiensis-triradiatus* Zone; S—*stephanephorus* Zone.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

dated Windsor Group strata of late Asbian age, and overlying palynologically constrained beds of latest Brigantian to Pendleian age, justify the assignment of the Middleborough and Maringouin Formations to the Brigantian. This defines the lower part of our magnetic polarity stratigraphic compilation, albeit without rigorous definition of the base of the Brigantian. Clearly, more paleomagnetic sampling of Windsor Group strata is needed to resolve the magnetostratigraphy within that group, and concurrently, to document the position of the *Knoxisporites stephanephorus–Schopfipollenites acadiensis–Knoxisporites triradiatus* concurrent range zone boundary in biologically constrained magnetostratigraphic sections. Our efforts to accomplish this goal during the present investigations were unsuccessful.

New polarity stratigraphy for the basal Pennsylvanian on Cape Breton Island appears to fill a gap in the Joggins section polarity pattern reported by DiVenere and Opdyke (1991b). This sampling gap spans in large part the depositional record of the Boss Point Formation at Joggins, where drab channel sandstones with intercalated gray mudrocks failed to provide suitable rock types for magnetic stratigraphy. We show here that the thick unsampled lower Boss Point interval in Di Venere and Opdyke (1991b) is the probable biostratigraphic equivalent of the lower Port Hood Formation in western Cape Breton Island, in which we find dominant normal polarity. Thick, drab multistoried channel sandstones characterize both the Boss Point and lower Port Hood Formations, but they differ in the associated overbank strata, which are mainly gray in the Boss Point Formation and red (subaerially oxidized) in the lower Port Hood Formation. We suggest that these rock units are transitional laterally one with the other, just as the Boss Point Formation at Joggins is transitional upward into red beds of the Little River Formation. These new data reconcile the work of Di Venere and Opdyke (1991b) with the results of Buchan and Chandler (1999), who first documented widespread indications of normal polarity in red strata conformably overlying typical Boss Point Formation channel sandstones in the New Glasgow area of Nova Scotia. We note that the reversed polarity interval low in the Boss Point Formation (Di Venere and Opdyke, 1991b), within which the thin normal magnetozone N8 occurs, has no apparent representation in the Cape Breton Island sections. It is possible that the reversed polarity interval is hidden in sampling gaps in the latter sections. We suggest, alternatively, that the basal Boss Point Formation may in fact correlate to the top of the normal zone preceding the Kiaman and is slightly younger than the basal Port Hood Formation, albeit palynologically indistinguishable, thereby explaining the absence of this reversed polarity magnetozone and normal zone N8 at Broad Cove Chapel.

Upper portions of the Port Hood Formation at the Broad Cove Chapel section are dominated by reverse polarity, comparable to post–Boss Point Formation strata at the Joggins section studied by Di Venere and Opdyke (1991b) and most recently assigned to the Little River Formation (Calder et al., 2005). We correlate the level of this normal-to-reversed polarity change within the lower Port Hood Formation to the base of the Kiaman reversed superchron (Irving and Parry, 1963). We cannot be more precise than Yeadonian–Langsettian for the age of this polarity change, but we note that the work of Nick et al. (1991) in concert with the biostratigraphic efforts of Spencer and Ross (1997) together establish evidence for normal polarity of Langsettian age within the Black Prince Limestone in Arizona and New Mexico, USA. An early Langsettian age for the base-Kiaman is thus supported (cf. Opdyke et al., 2000).

Correlation to the United States

Our correlations suggest that the youngest beds of the Mauch Chunk Formation represent a position relatively low in the Arnsbergian stage compared to the youngest strata in the eastern Canadian Arnsbergian sections. These correlations suggest a Brigantian assignment for the lower Mauch Chunk, supporting the assessment of Maynard et al. (2006), typified by alternating normal and reverse polarity as recorded in the Middleborough and Maringouin Formations in eastern Canada. We equate magnetozone MMCn (Opdyke and Di Venere, 2004) with magnetozone N5 in eastern Canada (cf. Di Venere and Opdyke, 1991a, 1991b), thus correlating the upper parts of the Mauch Chunk, characterized by both reverse and normal polarity, with the late Pendleian–early Arnsbergian parts of the Canadian sections. In the Pottsville and Lavelle sections where the sediments directly below the Mississippian-Pennsylvanian boundary unconformity have been sampled, the sediments have normal polarities. No samples were studied in Pennsylvanian-aged rocks above the Mauch Chunk red sediments (Fig. 11) because they are dominated by drab-colored sediments, many of which are very coarse and unsuitable for magnetic polarity study. A regional disconformity at the Mississippian-Pennsylvanian boundary has been postulated by several workers, most recently by Maynard et al. (2006, their fig. 3), who asserted that the Mauch Chunk is not older than Brigantian and not younger than approximately mid-Arnsbergian, and is separated from the overlying Pocahontas Formation by a significant unconformity. The Pocahontas Formation is of Namurian B (Marsdenian) age based on palynology (Peppers, 1996), or latest Namurian (Yeadonian) age (Blake, 2006) based on macroflora, and Blake and Beuthin (2008) identified the Mississippian-Pennsylvanian boundary interval within the Central Appalachian Basin as a significant disconformity. In Blake and Beuthin's (2008) estimate, following Blake (2006) and based wholly on macrofloral elements, strata beneath the unconformity in that area (Bluestone Formation) were of Arnsbergian but not latest Arnsbergian age, and strata immediately above the unconformity (i.e., the Pocahontas Formation) are youngest Namurian (Yeadonian) in age, ranging into the Langsettian. Thus, combining the work of Blake (2006), with that of Maynard et al. (2006) and Blake and Beuthin (2008), we identify a mid-Arnsbergian to Yeadonian hiatus at the Mississippian-Pennsylvanian boundary in the Central Appalachian Basin of the United States. It should be noted that the magnetic polarity stratigraphy in the central Appalachians is entirely from the type area of the Mauch Chunk in Pennsylvania. Our assessment of the unconformity in the United States described here is in reasonable agreement with our interpretation of the same boundary in the Nova Scotia–New Brunswick border region in eastern Canada, where we propose that the hiatus extends from early Arnsbergian to Yeadonian. However, the Cape Breton Island sections seem to preserve significantly more of the Arnsbergian than the Joggins section, suggesting that the apparent duration of the hiatus at the Mississippian-Pennsylvanian unconformity may in some areas be amplified by erosion and downcutting into underlying Arnsbergian strata. The Mississippian-Pennsylvanian unconformity in the United States shows similar evidence for major stratigraphic incision in the Central Appalachian Basin, where Pennsylvanian rocks may lie on rocks as old as the Brigantian (according to Maynard et al., 2006) Bluefield Formation (Blake and Beuthin, 2008).

The correlation of polarity zones that we propose in this paper, linking Mississippian sections more than 1500 km apart, is based on a pattern of magnetic polarity reversals spanning some 12 m.y. of late Mississippian and earliest Pennsylvanian time (Fig. 10). During the Brigantian substage, reverse polarity chrons dominate intervals of normal polarity at a ratio of ~2:1. In the early Namurian, Pendleian substage, intervals of normal and reversed polarity are in approximately equal proportion overall, but one long normal polarity chron (N5 or MMCn) makes a useful marker. In the Arnsbergian substage, eastern Canadian sections are

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

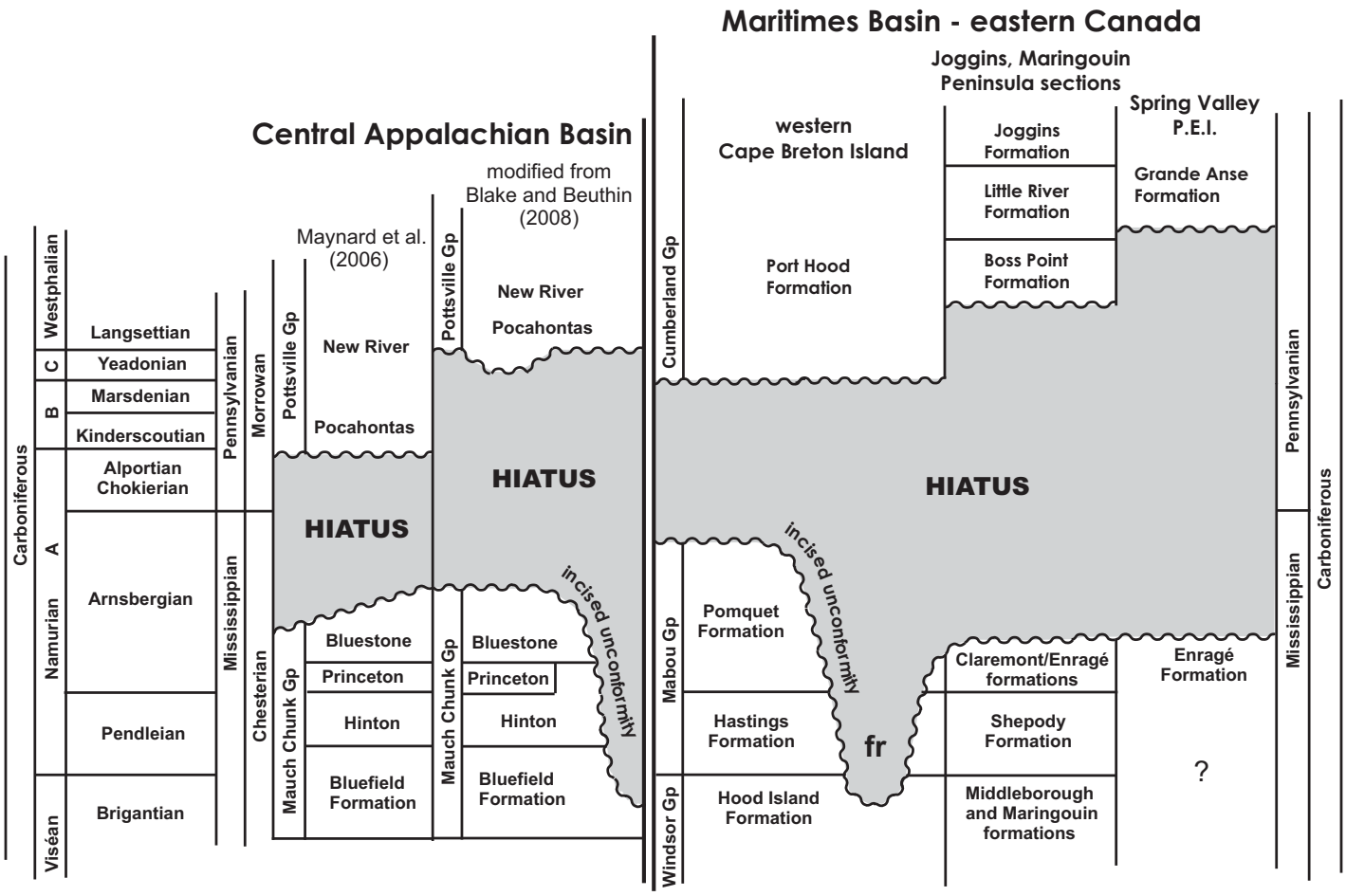


Figure 11. A comparison of the Mississippian-Pennsylvanian boundary unconformity in the Central Appalachian Basin with the Maritimes Basin of eastern Canada. P.E.I.—Prince Edward Island; fr—field relationships.

dominated by normal polarity. The uppermost Arnsbergian is, to our knowledge, not preserved in eastern North American sections.

North American magnetic polarity reversal patterns for the last 12 m.y. of Mississippian time are becoming more and more secure (Fig. 10). Published reconnaissance work has shown that both reversed and normal polarity intervals exist in Tournaisian and earlier Viséan sections, indicating that a very real potential exists for successful documentation of polarity patterns within virtually the entire Mississippian interval. A significant challenge to early Pennsylvanian magnetic polarity stratigraphy in North America is the apparent gap in polarity data for the Chokierian, Alportian, Kinderscoutian, and Marsdenian substages of the Bashkirian.

CONCLUSIONS

The new magnetic polarity stratigraphy from the middle Carboniferous is based on a high-temperature "C" component isolated above 600 °C that is shown to be prefolding. The resulting magnetic polarity data were therefore acquired early in the history of the sedimentary rocks and can be used to create a magnetic polarity stratigraphy.

New palynological studies place the age of the magnetic polarity stratigraphy in sedimentary rocks to range from late Asbian (late Mississippian) to the early Pennsylvanian *fulva* zone. The magnetic polarity of the uppermost Mississippian strata is normal in both Cape Breton Island and western Nova Scotia and New Brunswick. The stratigraphy of the region is complicated by a major disconformity separating the Mississippian and Pennsylvanian strata. The substages missing at the unconformity are mainly of early Pennsylvanian age and are the Chokierian, Alportian, Kinderscoutian, and Marsdenian (Namurian A and B). Above the disconformity on Cape Breton Island, the earliest Pennsylvanian strata are all of normal polarity.

The base of the Kiaman has never before been identified in a sedimentary section. We propose that the beginning of the Kiaman is present in the Broad Cove Chapel section at 1280 m above the base of the section, where a change from normal to reversed polarity takes place. This position is constrained by previous work on the Little River Formation of Calder et al. (2005) by Di Venere and Opdyke (1991b), where only reversed polarity is found in rocks assigned to the same spore zone but slightly younger in age. The estimated age of this important horizon is 318 Ma using the new time scale of Gradstein et al. (2012). This date would make the Kiaman ~58 m.y. in duration.

The new palynologic data from the Carboniferous of New Brunswick and Nova Scotia have enabled us to propose robust magnetic polarity stratigraphy correlations between the Maritimes Basin and Pennsylvania (Fig. 10). The lithostratigraphy, biostratigraphy, and magnetic polarity stratigraphy for the time interval are given in Figure 12. We have numbered the polarity sequence using the designated age beginning with the oldest, and using the age (substage)

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

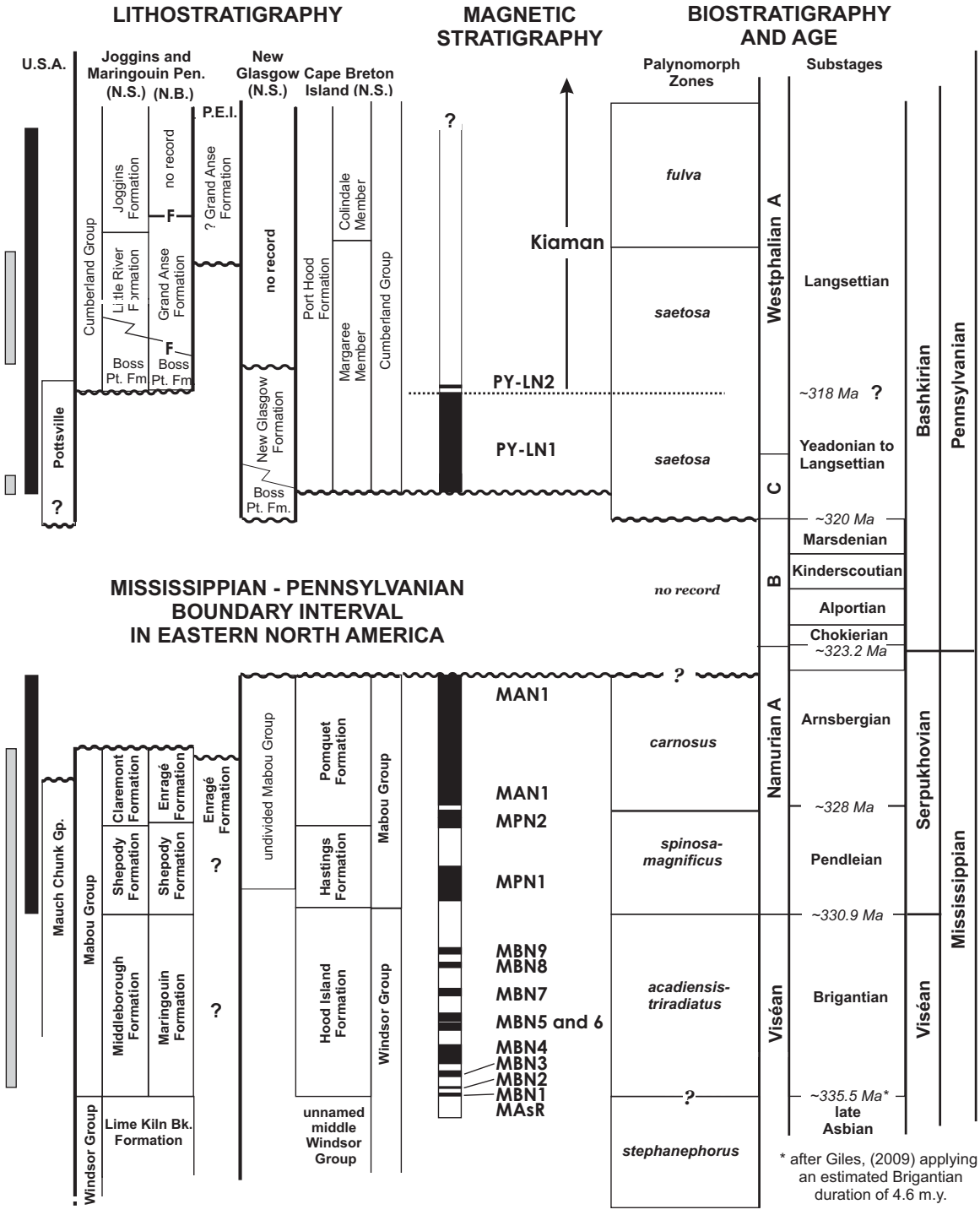


Figure 12. Late Mississippian and early Pennsylvanian lithostratigraphy, biostratigraphy, and magnetic polarity stratigraphy of eastern Canada. "F"—faulted contact; N.S.—Nova Scotia; N.B.—New Brunswick; P.E.I.—Prince Edward Island. Black bar on the left illustrates the stratigraphic range of new biostratigraphic and magnetostratigraphic results. Gray bar at the left shows the stratigraphic range of previously reported data.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

designation as a prefix. The oldest magnetozone seen in the studied sections is Asbian in age and has reverse magnetization; we therefore designate it as MAsR, for Mississippian, Asbian, reversed chron. Higher in the stratigraphic column, the succeeding normal zone is designated MBN1, for Mississippian, Brigantian, normal, 1. We have applied analogous procedures upward through the stratigraphic column. The oldest Pennsylvanian magnetozone is designated PY-LN1 (Pennsylvanian, Yeadonian to Langsettian, normal, 1). The use of local terminology should be terminated for correlating sections that are widely separated, although independent biostratigraphic support for these correlations is clearly needed.

ACKNOWLEDGMENTS

We would like to thank Lisa Tauxe, who was of great assistance with Figure 4. Randy Enkin of Geological Survey of Canada (Pacific) kindly loaned sampling and core orientation devices for our use. Clinton St. Peter and Susan Johnson shared unpublished logs of the section near Green Creek on the Maringouin Peninsula. Sonya Dehler and Alan Grant, Geological Survey of Canada (Atlantic), reviewed the manuscript and offered suggestions for improvement. The constructive reviews of John Geissman and Ken Kodama are much appreciated. The paper is Earth Science Sector (ESS) contribution 20100502.

REFERENCES CITED

Belt, E.S., 1964, Revision of Nova Scotia middle Carboniferous units: American Journal of Science, v. 262, p. 653–673, doi:10.2475/ajs.262.5.653.

Belt, E.S., 1965, Stratigraphy and paleogeography of Mabou Group and related middle Carboniferous facies, Nova Scotia, Canada: Geological Society of America Bulletin, v. 76, p. 777–802, doi:10.1130/0016-7606(1965)76[777:SAPOMG]2.0.CO;2.

Bilardello, D., and Kodama, K.P., 2009, Palaeomagnetism and magnetic anisotropy of Carboniferous redbeds from the Maritime Provinces of Canada: Evidence for shallow palaeomagnetic inclinations and implications for North American apparent polar wander: Geophysical Journal International, v. 180, p. 1013–1029, doi:10.1111/j .1365-246X.2009.04457.x.

Blake, B.M., 2006, Macrofloral evidence for an unconformity at the mid-Carboniferous boundary in the Central Appalachian Basin: Geological Society of America Abstracts with Programs, v. 38, no. 7, p. 317.

Blake, B.M., and Beuthin, J.D., 2008, Deciphering the mid-Carboniferous eustatic event in the central Appalachian foreland basin, southern West Virginia, USA, *in* Fielding, C.R., Frank, T.D., and Isbell, J.L., eds., Resolving the Late Paleozoic Ice Age in Time and Space: Geological Society of America Special Paper 441, p. 249–260, doi:10.1130/2008.2441(17).

Boehner, R.C., 1985, Mabou Group, *in* Williams, G.L., Fyffe, L.R., Wardle, R.J., Coleman-Sadd, S.P., Boehner, R.C., and Watt, J.A., eds., Lexicon of Canadian Stratigraphy: Volume VI. Atlantic Region: Calgary, Canadian Society of Petroleum Geologists, p. 223–224.

Boehner, R.C., and Giles, P.S., 1982, Geology of the Antigonish Basin, Antigonish County, Nova Scotia: Nova Scotia Department of Natural Resources, Mines and Energy Branch, Memoir 8, 109 p.

Buchan, K.L., and Chandler, F.W., 1999, Paleomagnetism of the distal member of the New Glasgow Formation, Nova Scotia, Canada: New constraints on magnetic polarity stratigraphy near the base of the Kiaman superchron: The Journal of Geology, v. 107, p. 271–286, doi:10.1086/314347.

Calder, J.H., Rygel, M.C., Ryan, R.J., Falcon-Lang, H.J., and Hebert, B.L., 2005, Stratigraphy and sedimentology of early Pennsylvanian redbeds at Lower Cove, Nova Scotia, Canada: The proposed Little River Formation with redefinition of the Joggins Formation: Atlantic Geology, v. 41, p. 143–167.

Davies, S.J., Gibling, M.R., Rygel, M.C., and Calder, J.H., 2005, The Pennsylvanian Joggins Formation of Nova Scotia: Sedimentological log and stratigraphic framework of the historic fossil cliffs: Atlantic Geology, v. 41, p. 115–142.

Di Venere, V.J., and Opdyke, N.D., 1990, Paleomagnetism of the Maringouin and Shepody formations, New Brunswick: A Namurian magnetic stratigraphy: Canadian Journal of Earth Sciences, v. 27, p. 803–810, doi:10.1139/e90-082.

Di Venere, V.J., and Opdyke, N.D., 1991a, Magnetic polarity stratigraphy in the uppermost Mississippian Mauch Chunk Formation, Pottsville, Pennsylvania: Geology, v. 19, p. 127–130, doi:10.1130/0091-7613(1991)019<0127:MPSITU>2.3.CO;2.

Di Venere, V.J., and Opdyke, N.D., 1991b, Magnetic polarity stratigraphy and Carboniferous paleopole positions from the Joggins section, Cumberland Basin, Nova Scotia: Journal of Geophysical Research, v. 96, p. 4051–4064, doi:10.1029/90JB02148.

Fisher, R.A., 1953, Dispersion on a sphere: Proceedings of the Royal Society of London, ser. A, Mathematical and Physical Sciences, v. 217, p. 295–305, doi:10.1098/rspa.1953.0064.

Gibling, M.R., Culshaw, N., Rygel, M.C., and Pascucci, V., 2009, The Maritimes Basin of Atlantic Canada: Basin creation and destruction in the collisional zone of Pangea, *in* Miall, A.D., ed., The Sedimentary Basins of the United States and Canada, Volume 5: Amsterdam, Elsevier, p. 211–244.

Giles, P.S., 2009, Orbital forcing and Mississippian sea level change: Time series analysis of marine flooding events in the Viséan Windsor Group of eastern Canada and implications for Gondwana glaciations: Bulletin of Canadian Petroleum Geology, v. 57, p. 1–23.

Giles, P.S., and Utting, J., 1999, Maritimes Basin Stratigraphy—Prince Edward Island and Adjacent Gulf of St. Lawrence: Geological Survey of Canada Open-File 3732, Well Correlation Panel.

Giles, P.S., Allen, T.L., and Hein, F.J., 1997a, Bedrock Geology of Cheticamp Map-Area (11K11, 11K10), Cape Breton Island, Nova Scotia: Geological Survey of Canada Open-File 3455, scale 1:50,000 map, with marginal notes.

Giles, P.S., Hein, F.J., and Allen, T.L., 1997b, Bedrock Geology of Margaree (11K06), Cape Breton Island, Nova Scotia: Geological Survey of Canada Open-File 3254, scale 1:50,000 map, with marginal notes.

Giles, P.S., Hein, F.J. and Allen, T.L., 1997c, Bedrock Geology of Port Hood–Lake Ainslie (11K04, 11K03, 11F13), Cape Breton Island, Nova Scotia: Geological Survey of Canada Open-File 3253, scale 1:50,000 map, with marginal notes.

Gradstein, F.M., Ogg, J.G., Schmitz, M., and Ogg, G., 2012, The Geological Timescale 2012: Amsterdam, Elsevier, 1176 p., doi:10.1016/B978-0-444-59425-9.00002-0.

Hamblin, A.P., 2001, Stratigraphy, Sedimentology, Tectonics, and Resource Potential of the Lower Carboniferous Mabou Group, Nova Scotia: Geological Survey of Canada Bulletin 568, 166 p.

Irving, E., and Parry, L.G., 1963, The magnetism of some Permian rocks from New South Wales: Geophysical Journal of the Royal Astronomical Society, v. 7, p. 395, doi:10.1111/j.1365-246X.1963.tb07084.x.

Keighley, D.G., and Pickerill, R.K., 1996, The evolution of fluvial systems in the Port Hood Formation (Upper Carboniferous), western Cape Breton Island, eastern Canada: Sedimentary Geology, v. 106, p. 97–144, doi:10.1016/0037-0738(95)00154-9.

Kent, D.V., Olsen, P.E., and Witte, W.K., 1995, Late Triassic–earliest Jurassic geomagnetic polarity sequence and paleolatitudes from drill cores in the Newark rift basin, eastern North America: Journal of Geophysical Research, v. 100, no. B8, p. 14,965–14,998, doi:10.1029/95JB01054.

King, R.F., 1955, The remanent magnetism of artificially deposited sediments: Geophysical Journal International, v. 7, p. 115–134, doi:10.1111/j.1365-246X.1955.tb06558.x.

Kirschvink, J.L., 1980, The least squares line and plane and the analysis of paleomagnetic data: Journal of the Royal Astronomical Society, v. 62, p. 699–718, doi:10.1111/j.1365-246X.1980.tb02601.x.

Kodama, K.P., 2012, Paleomagnetism of Sedimentary Rocks: Process and Interpretation: Wiley-Blackwell, 164 p.

Maynard, J.P., Eriksson, K.A., and Law, R.D., 2006, The Upper Mississippian Bluefield Formation in the Central Appalachian Basin: A hierarchical sequence-stratigraphic record of a greenhouse to icehouse transition: Sedimentary Geology, v. 192, p. 99–122, doi:10.1016/j.sedgeo.2006.03.027.

Menning, M., Alekseev, A.S., Chuvashov, B.I., Davydov, V.I., Devuyst, F.-X., Forke, H.C., Grunt, T.A., Hance, L., Heckel, P.H., Izokh, N.G., Jin, Y.-G., Jones, P.J., Kotlyar, G.V., Kozur, H.W., Nemyrovska, T.I., Schneider, J.W., Wang, X.-D., Weddige, K., Weyer, D., Work, D.M., 2006, Global time scale and regional stratigraphic reference scales of Central and West Europe, East Europe, Tethys, South China, and North America as used in the Devonian–Carboniferous–Permian Correlation Chart 2003 (DCP 2003): Palaeogeography, Palaeoclimatology, Palaeoecology, v. 240, p. 318–372.

Neves, R., and Belt, E.S., 1971, Some observations of Namurian and Viséan spores from Nova Scotia, Britain and northern Spain, *in* Sixième Congrès International de Stratigraphie et de Géologie du Carbonifère, Sheffield, 1967: Compte Rendu, v. 3, p. 1233–1249.

Nick, K., Xia, K., and Elmore, R.D., 1991, Paleomagnetic and petrographic evidence for early magnetizations in successive terra rosa paleosols, lower Pennsylvanian Black Prince limestone, Arizona: Journal of Geophysical Research, v. 96, no. B6, p. 9873–9885, doi:10.1029/91JB00338.

Opdyke , N.D., and Channell, J.E.T., 1996, Magnetic Stratigraphy: New York and London, Academic Press, 346 p.

Opdyke, N.D., and Di Venere, V.J., 2004, The magnetic polarity stratigraphy of the Mauch Chunk Formation, Pennsylvania: Proceedings of the National Academy of Sciences of the United States of America, v. 101, p. 13,423–13,427, doi:10.1073/pnas.0403786101.

Opdyke, N.D., Roberts, J., Claoué-Long, J., Irving, E., and Jones, P.J., 2000, Base of the Kiaman: Its definition and global stratigraphic significance: Geological Society of America Bulletin, v. 112, p. 1315–1341, doi:10.1130/0016-7606(2000)112<1315:BOTKID>2.0.CO;2.

Roy, J.L., and Park, J.K., 1969, Paleomagnetism of the Hopewell Group, New Brunswick: Journal of Geophysical Research, v. 74, p. 594–604, doi:10.1029/JB074i002p00594.

Roy, J.L., and Robertson, W.A., 1968, Evidence for diagenetic remanent magnetization in the Maringouin Formation: Canadian Journal of Earth Sciences, v. 5, p. 275–285, doi:10.1139/e68-029.

Ryan, R.J., and Boehner, R.C., 1994, Geology of the Cumberland Basin, Cumberland, Colchester and Pictou Counties, Nova Scotia: Nova Scotia Department of Natural Resources Memoir 10, 222 p.

St. Peter, C.J., and Johnson, S.C., 2009, Stratigraphy and Structural History of the Late Paleozoic Maritimes Basin in Southeastern New Brunswick, Canada: New Brunswick Department of Natural Resources; Minerals, Policy and Planning Division, Memoir 3, 348 p.

Spencer, P.K., and Ross, C.A., 1997, Black Prince limestone and its foraminifers, Upper Mississippian–Lower Pennsylvanian, S.E. Arizona and S.W. New Mexico, *in* Ross, C.A., Ross, J.R.P., and Brenkle, P.L., eds., Late Paleozoic Foraminifera: Their Biostratigraphy, Evolution and Paleoecology; and the Mid-Carboniferous Boundary: Cushman Foundation for Foraminiferal Research Special Publication 36, p. 139–143.

Symons, D.T.A., 1990, Early Permian pole: Evidence from the Pictou red beds, Prince Edward Island, Canada: Geology, v. 18, p. 234–237, doi:10.1130/0091-7613(1990)018<0234:EPPEFT>2.3.CO;2.

Downloaded from gsabulletin.gsapubs.org on June 29, 2014

Tauxe, L., and Kent, D.V., 1984, Properties of a detrital remanence carried by hematite from study of modern river deposits and laboratory redeposition experiments: Geophysical Journal of the Royal Astronomical Society, v. 76, p. 543–561.

Tauxe, L., and Kent, D.V., 2004, A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: Was the ancient field dipolar?, *in* Channell, J.E.T., Kent, D.V., Lowrie, W., and Meert, J.G., eds., Timescales of the Paleomagnetic Field: American Geophysical Union Geophysical Monograph 145, p. 101–115.

Utting, J., 1980, Palynology of the Windsor Group (Mississippian) in a borehole at Stewiacke, Shubenacadie Basin, Nova Scotia: Canadian Journal of Earth Sciences, v. 17, p. 1031–1045, doi:10.1139/e80-103.

Utting, J., and Giles, P.S., 2004, Biostratigraphical implications of new palynological data from the Mississippian of Newfoundland and Nova Scotia, *in* Laurie, J.R., and Foster, C.B., eds. Palynological and Micropalaeontological Studies in Honour of Geoffrey Playford: Association of Australasian Palaeontologists Memoir 29, p. 115–160.

Utting, J., and Giles, P.S., 2008, Palynostratigraphy and lithostratigraphy of Carboniferous Upper Codroy Group and Barachois Group, southwestern Newfoundland: Canadian Journal of Earth Sciences, v. 45, p. 45–67, doi:10.1139/E07-066.

Utting, J., and Wagner, R.H., 2005, Megaflora of the Upper Carboniferous Cumberland Group, Joggins area, Nova Scotia and New Brunswick, Canada: Geological Society of Canada–Mineralogical Society of Canada–Canadian Society of Petroleum Geologists–Canadian Society of Soil Science, Joint Meeting, Halifax, Nova Scotia, Abstracts with Programs, p. 197.

Utting, J., Giles, P.S., Dolby, G., and Calder, J.H., 2005, New palynological and microfloral data from upper Viséan, Namurian and Westphalian rocks of Joggins area, Nova Scotia and New Brunswick, Canada: Geological Society of Canada–Mineralogical Society of Canada–Canadian Society of Petroleum Geologists–Canadian Society of Soil Science, Joint Meeting, Halifax, Nova Scotia, Abstracts with Programs, p. 197.

Utting, J., Giles, P.S., and Dolby, G., 2010, Palynostratigraphy of Mississippian and Pennsylvanian rocks, Joggins area, Nova Scotia and New Brunswick, Canada: Palynology, v. 34, p. 43–89, doi:10.1080/01916121003620569.

von Bitter, P.H., Giles, P.S., and Utting, J., 2006, Biostratigraphic correlation of major cycles in the Windsor and Codroy Groups of Nova Scotia & Newfoundland, Atlantic Canada, with the Mississippian substages of Britain and Ireland, *in* Wong, Th.E., ed., Proceedings of the XVth International Congress on Carboniferous and Permian Stratigraphy: Utrecht, the Netherlands, August 2003, Royal Netherlands Academy of Arts and Sciences, p. 513–534.

Wade, J.A., Grant, A.C., Sanford, B.V., and Barss, M.S., 1977, Basement Structure, Eastern Canada and Adjacent Areas: Geological Survey of Canada Map 1400A, 4 sheets, scale 1:2,000,000.

Science Editor: A. Hope Jahren
Associate Editor: Neil Tabor

Manuscript Received 18 June 2013
Revised Manuscript Received 30 January 2014
Manuscript Accepted 21 February 2014

Printed in the USA